

```
In [3]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

```
In [4]: #Task 1 Flipkart: Rating vs Price

df = pd.read_csv("flipkart.csv")
```

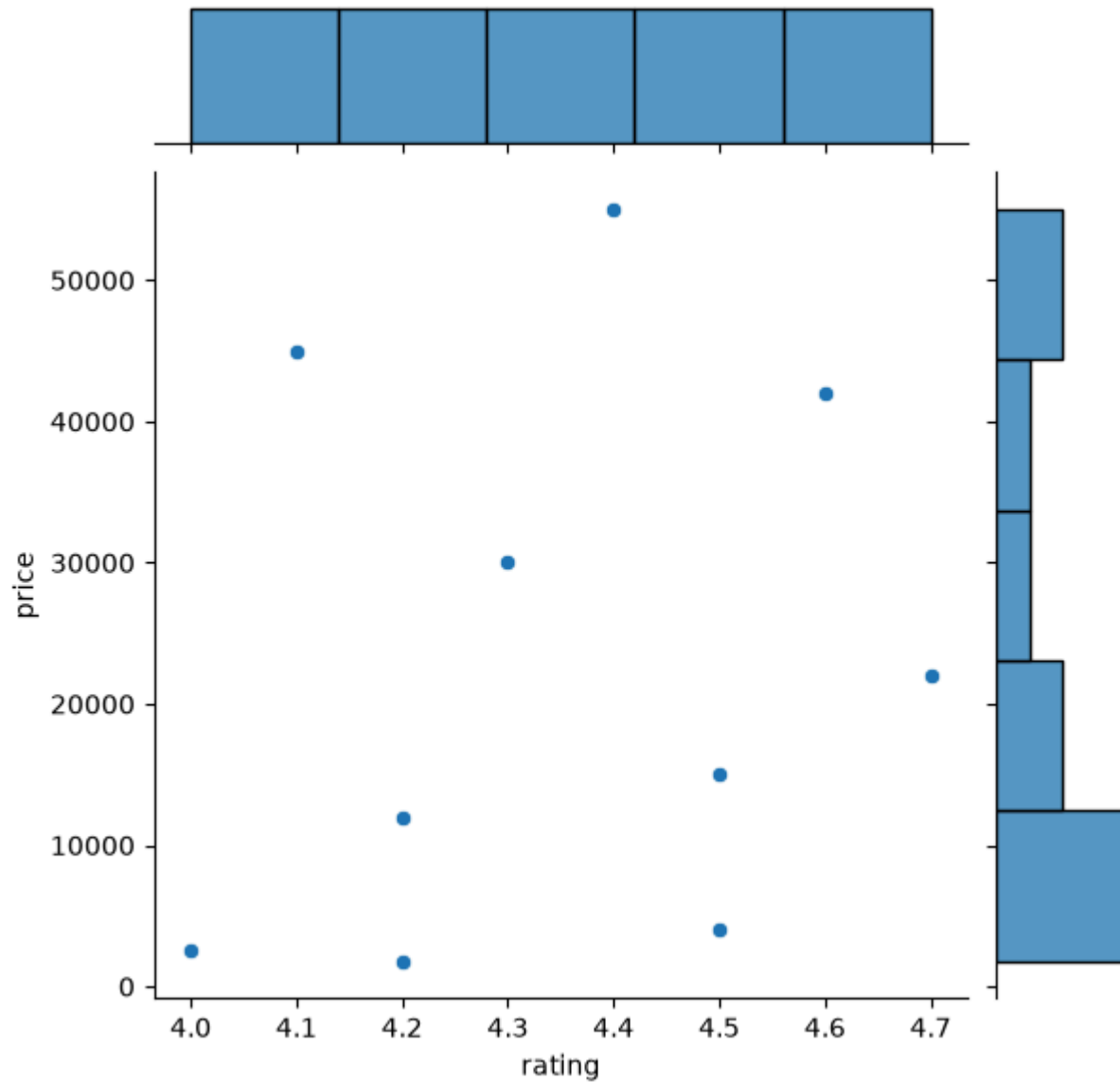
```
In [5]: df
```

```
Out[5]:
```

	product	rating	price
0	Phone A	4.5	15000
1	Phone B	4.2	12000
2	Phone C	4.7	22000
3	Laptop A	4.1	45000
4	Laptop B	4.4	55000
5	TV A	4.3	30000
6	TV B	4.6	42000
7	Watch A	4.0	2500
8	Watch B	4.5	4000
9	Earbuds A	4.2	1800

```
In [10]: sns.jointplot(x="rating", y="price", data=df, kind="scatter")
```

```
Out[10]: <seaborn.axisgrid.JointGrid at 0x266b559d450>
```



#The scatter jointplot shows the relationship between product rating and price. Each dot represents a product.

```
In [11]: #Task 2 – Swiggy: Delivery Time vs Order Value  
df = pd.read_csv("swiggy_orders.csv")
```

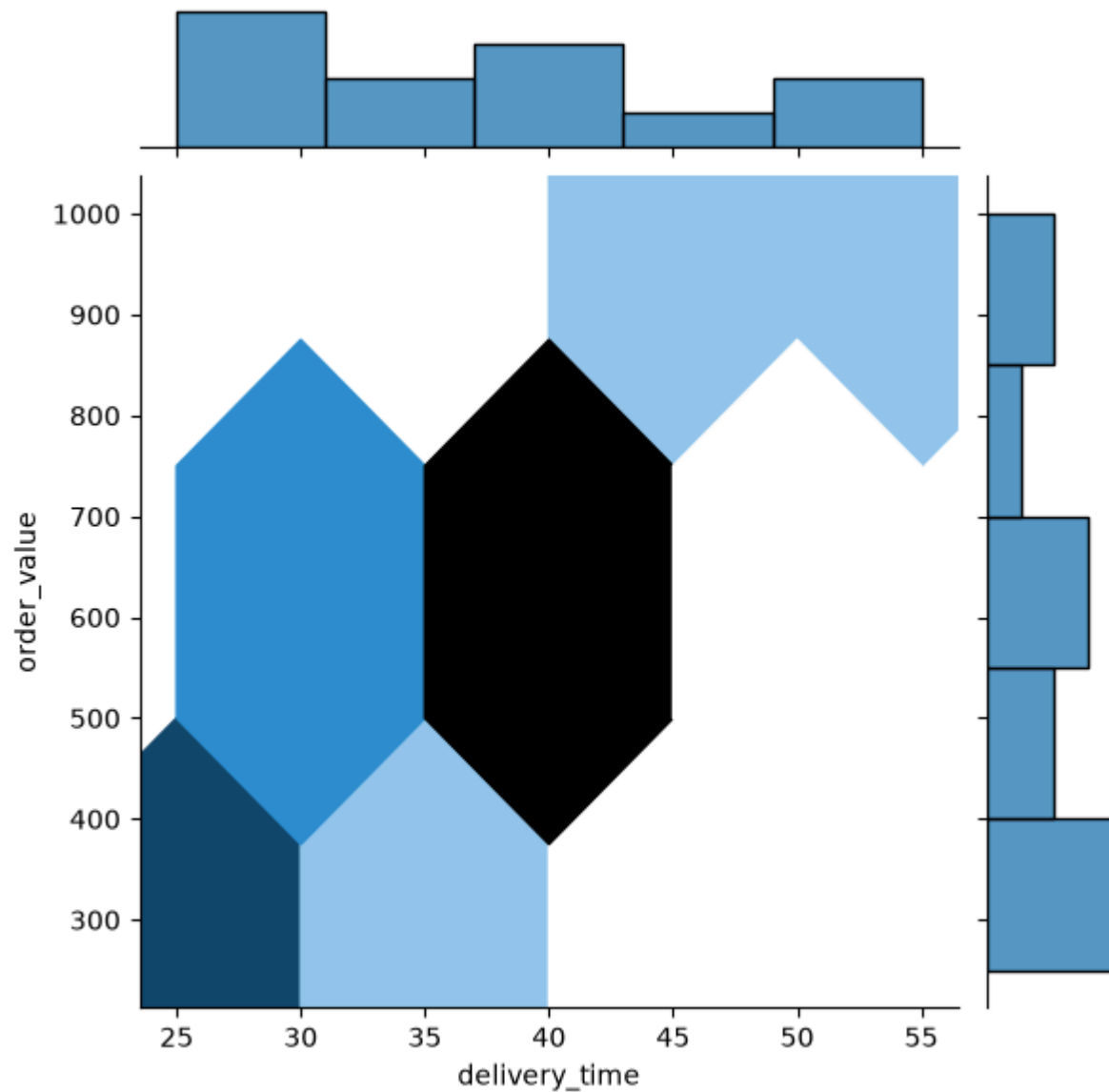
In [12]: df

Out[12]:

	order_id	delivery_time	order_value
0	101	25	250
1	102	30	400
2	103	35	350
3	104	40	600
4	105	28	300
5	106	45	750
6	107	50	900
7	108	32	450
8	109	38	550
9	110	55	1000
10	111	42	650
11	112	27	280

In [13]: sns.jointplot(x="delivery_time", y="order_value", data=df, kind="hex")

Out[13]: <seaborn.axisgrid.JointGrid at 0x266b30c29e0>



#The hex jointplot shows the relationship between delivery time and order value. It helps us understand where the orders are concentrated.

```
In [15]: #Task 3 – Instagram: Likes vs Comments
```

```
df = pd.read_csv("instagram_posts.csv")
```

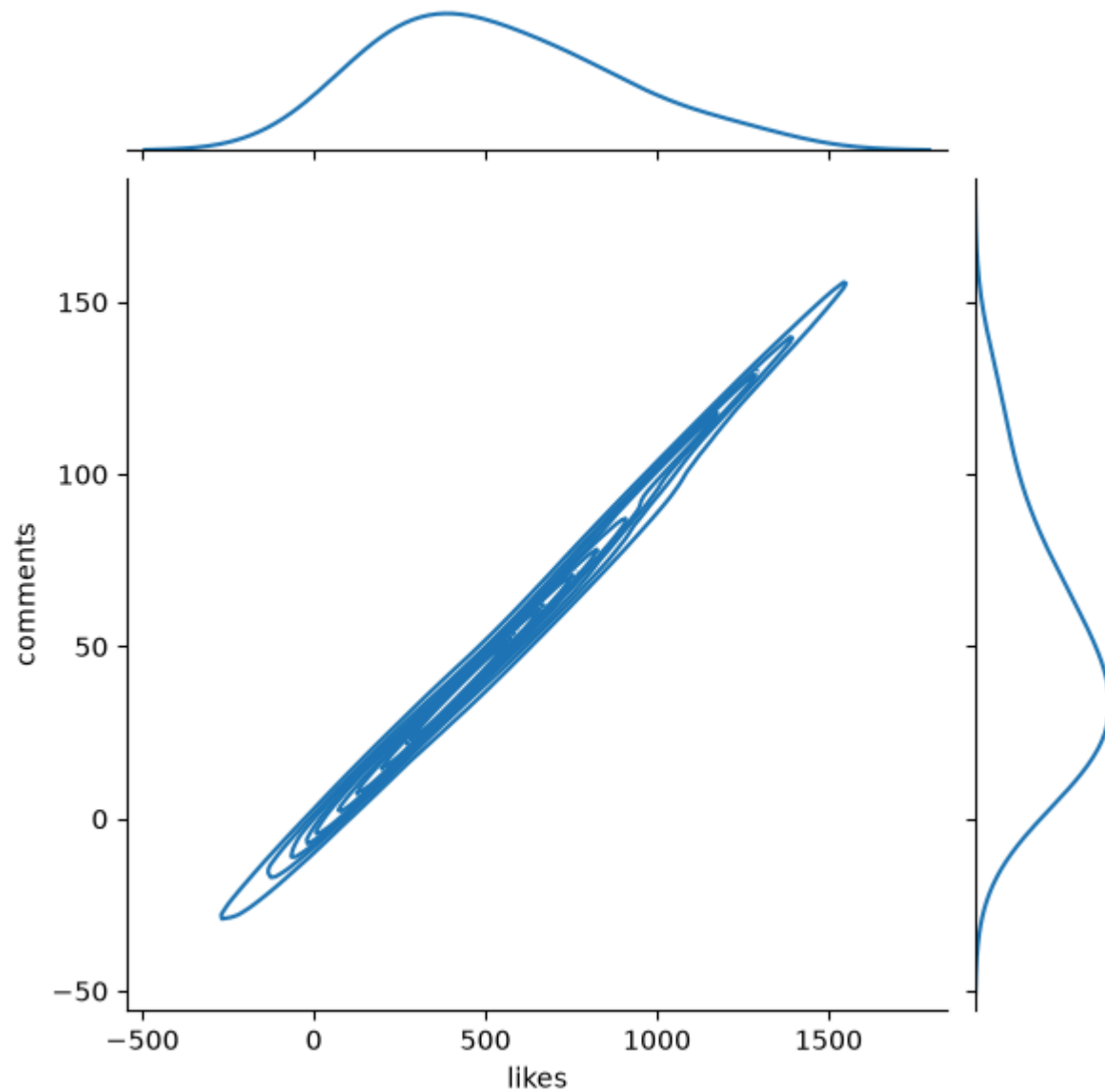
```
In [16]: df
```

```
Out[16]:
```

	post_id	likes	comments
0	1	100	10
1	2	250	20
2	3	180	15
3	4	500	45
4	5	700	60
5	6	320	25
6	7	900	90
7	8	450	40
8	9	1200	120
9	10	600	55
10	11	350	30
11	12	800	75

```
In [17]: sns.jointplot(x="likes", y="comments", data=df, kind="kde")
```

```
Out[17]: <seaborn.axisgrid.JointGrid at 0x266b30c1cd0>
```



The KDE jointplot shows the relationship between likes and comments. It shows the areas where Instagram posts are more concentrated.

```
In [18]: #Task 4 - Zomato: Rating vs Number of Reviews
```

```
df= pd.read_csv("zomato_reviews.csv")
```

In [19]:

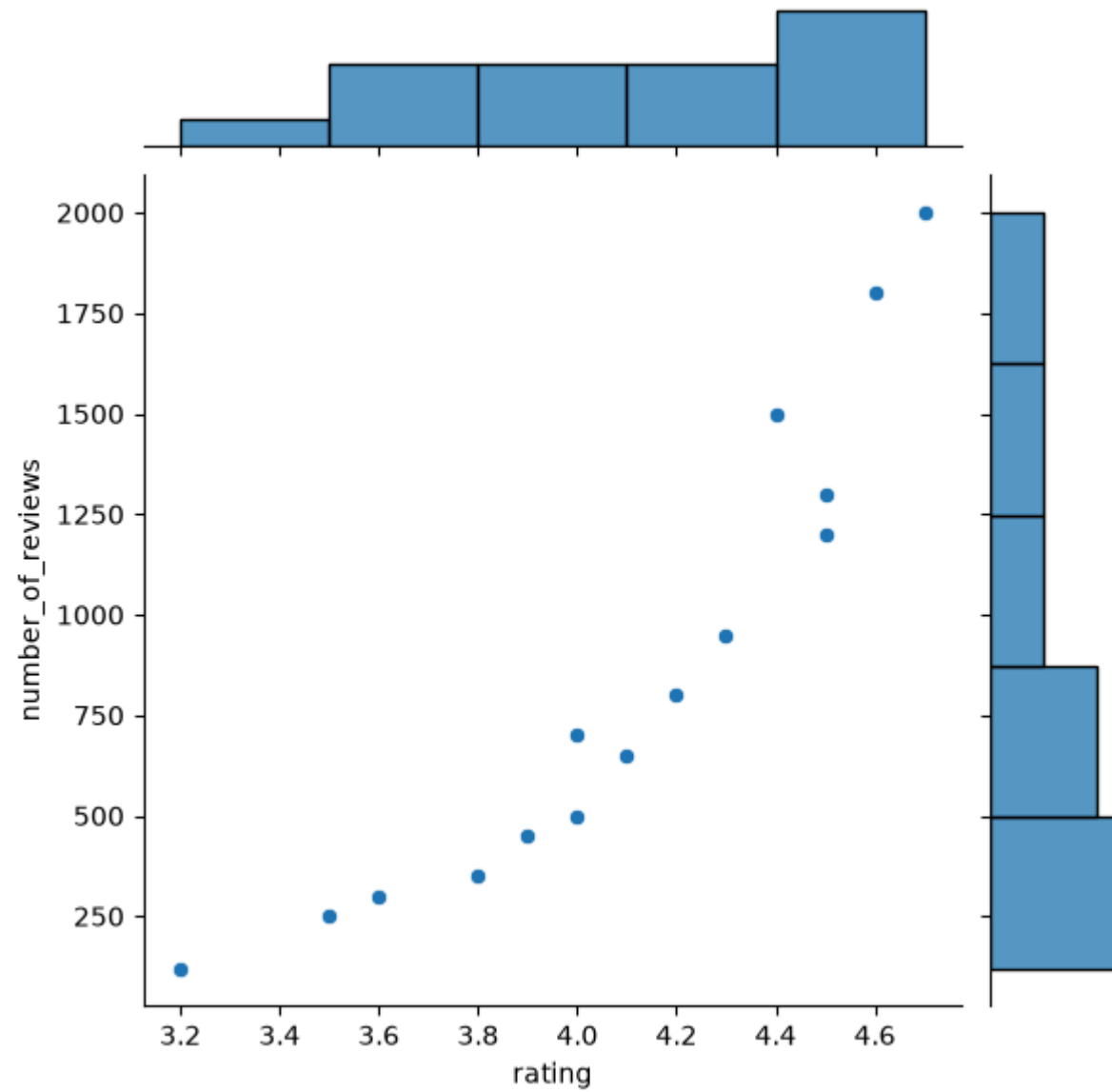
df

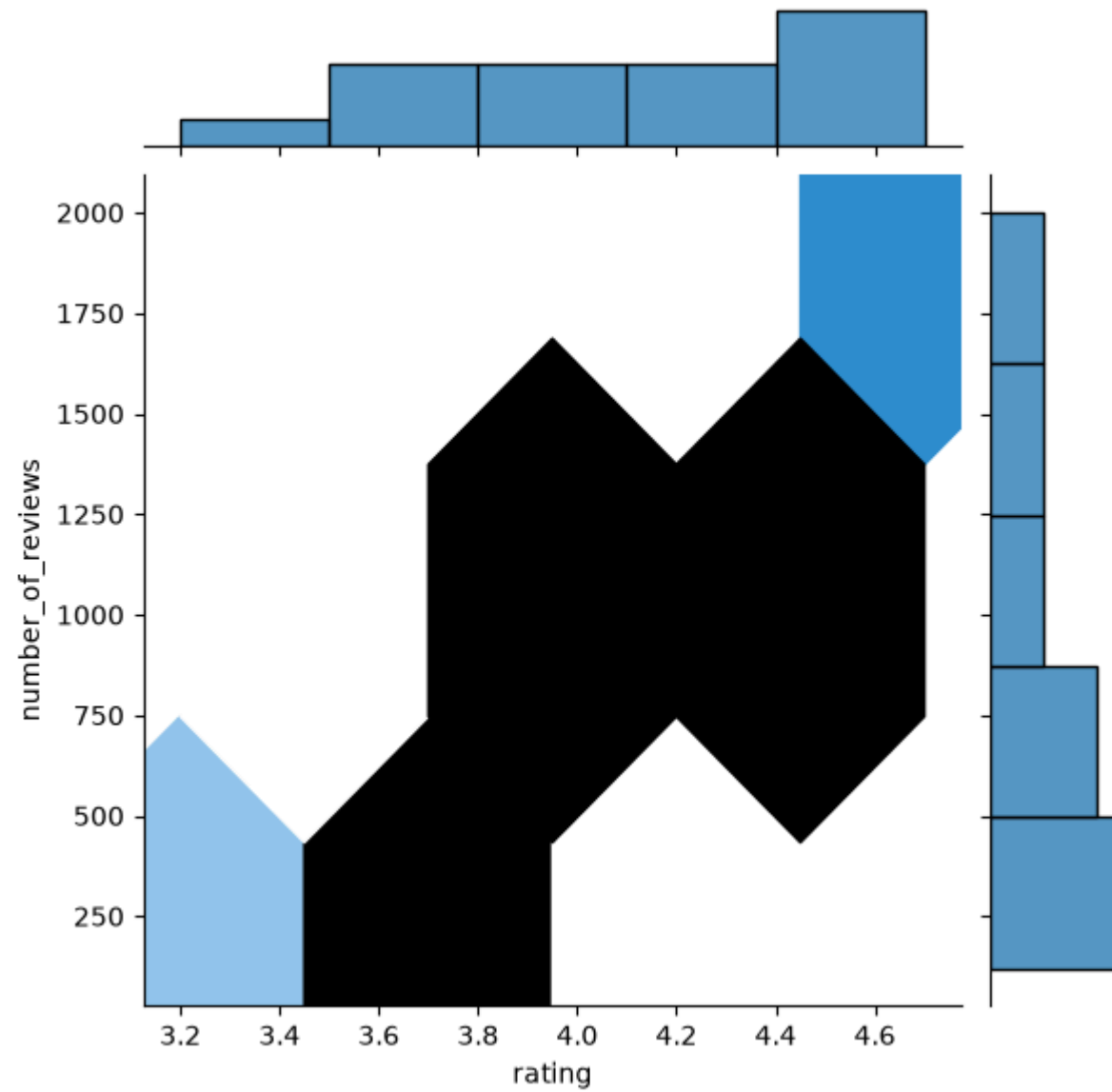
Out[19]:

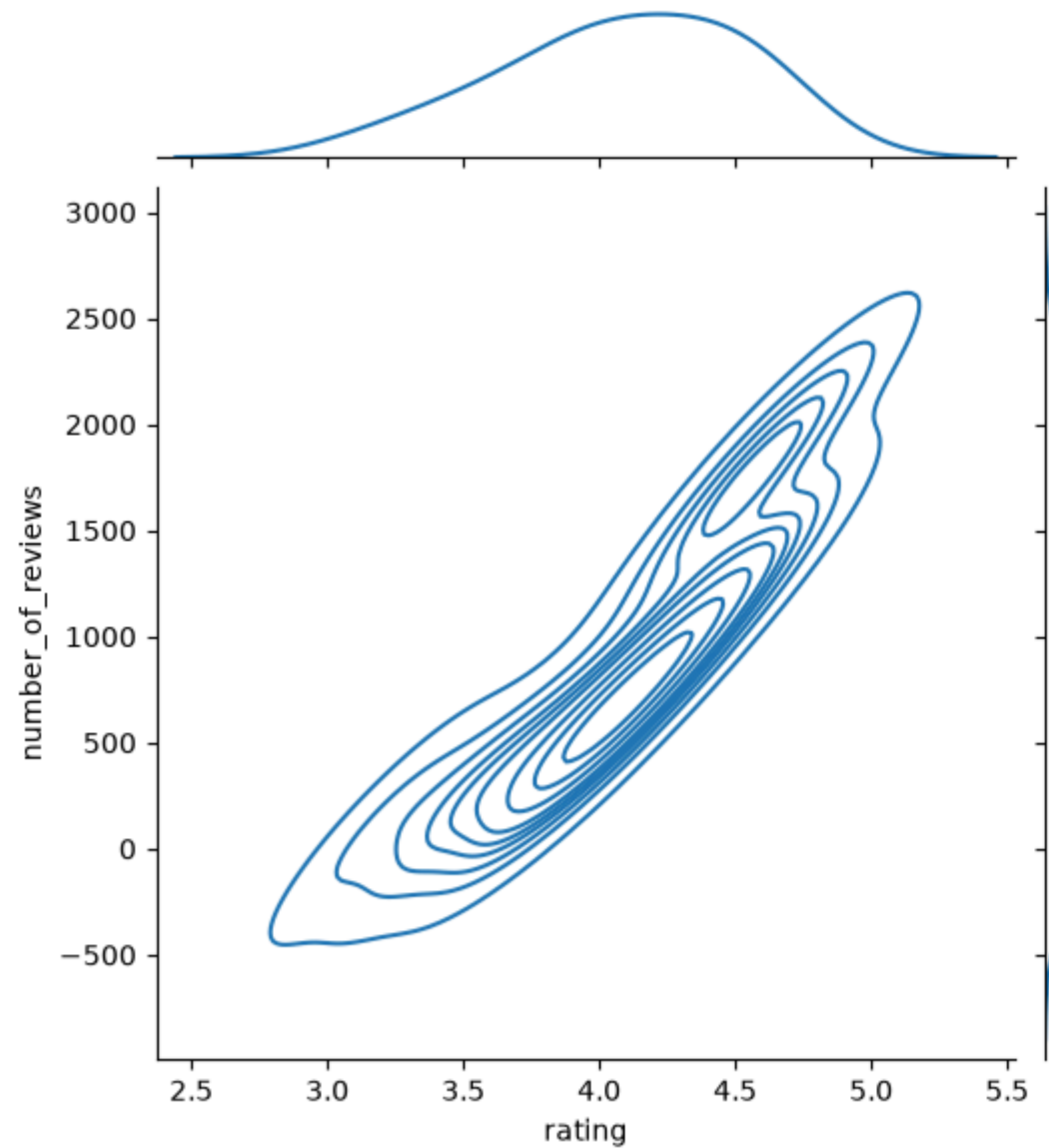
	restaurant	rating	number_of_reviews
0	A	3.2	120
1	B	3.5	250
2	C	4.0	500
3	D	4.2	800
4	E	4.5	1200
5	F	3.8	350
6	G	4.1	650
7	H	4.4	1500
8	I	3.6	300
9	J	4.7	2000
10	K	4.3	950
11	L	3.9	450
12	M	4.6	1800
13	N	4.0	700
14	O	4.5	1300

```
In [20]: sns.jointplot(x="rating", y="number_of_reviews", data=df, kind="scatter")
sns.jointplot(x="rating", y="number_of_reviews", data=df, kind="hex")
sns.jointplot(x="rating", y="number_of_reviews", data=df, kind="kde")
```

Out[20]: <seaborn.axisgrid.JointGrid at 0x266b3027ce0>







#Conclusion 1. Scatter plot shows individual restaurants as dots and helps us understand the relationship between rating and number of reviews. 2. Hex plot shows the concentration of restaurants using hexagonal areas. It is useful when there are many data points. 3. KDE plot shows the density of restaurants and helps us understand where most of the data is concentrated. For this dataset, the hex plot is useful for understanding data concentration because it shows where many data points are located without overlapping dots.